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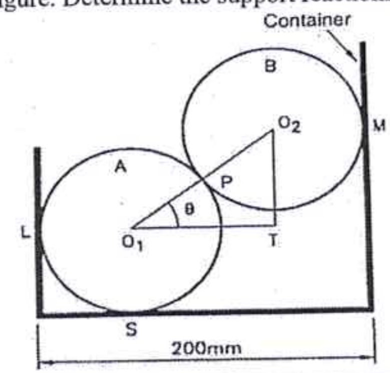
END SEMESTER EXAMINATION - 2024

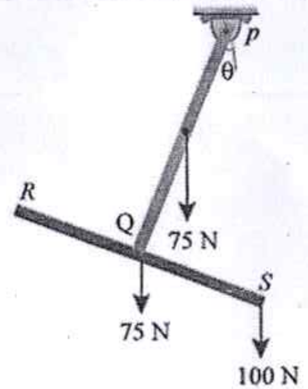
Name of the Course: B. Tech. Mech. — Semester: 3rd

Name of the Paper: Engineering Mechanics — Paper Code: TME 306

Time: 3 Hours — Maximum Marks: 100

Note:-
(i) All questions are compulsory.
(ii) Answer any two sub questions among a, b & c in each main question.
(iii) Total marks in each main question are twenty.
(iv) Each question carries 10 marks.

Q1	(20 marks)	
(a)	What are the methods for finding out the resultant force for a given system of forces? (discuss about different Law of forces with diagrams)	
(b)	The forces 20 N, 30 N, 40 N, 50 N and 60 N are acting at one of the angular points of a regular hexagon, towards the other five angular points, taken in order. Find the magnitude and direction of the resultant force.	**CO1**
(c)	Two spheres A and B weighing 100 N and 75 N respectively and with the corresponding radii 75 mm and 50 mm are placed in a container as shown in figure. Determine the support reactions.  	
Q2	(20 marks)	
(a)	A beam AB of length 5 m supported at A and B carries two point loads W_1 and W_2 of 3 kN and 5 kN which are 1 m apart. If the reaction at B is 2 kN more than that at A, find the distance between the support A and the load 3 kN.	**CO2**
(b)	Two identical prismatic bars PQ and RS each weighing 75 N are welded together to form a Tee and are suspended in a vertical plane as shown in Fig. Calculate the value of θ, that the bar PQ will make with vertical when a load of 100 N is applied at S.	

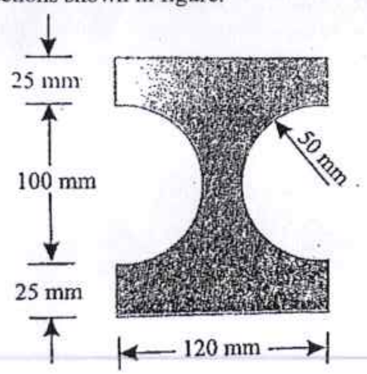


(c) i. State and prove Lami's Theorem
ii. State Varignon's Theorem
iii. State the condition of static equilibrium of a system.

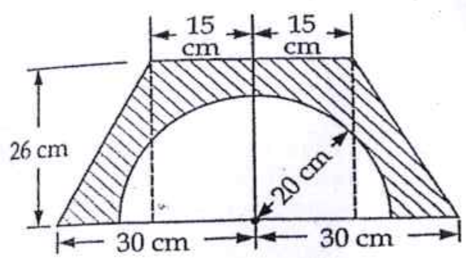
Q3 (20 marks) — CO3

(a) i. State the difference between centroid and centre of gravity.
ii. State and prove perpendicular axis of theorem. (3+7)

(b) Determine the horizontal and vertical centroidal moment of inertia of the sections shown in figure.

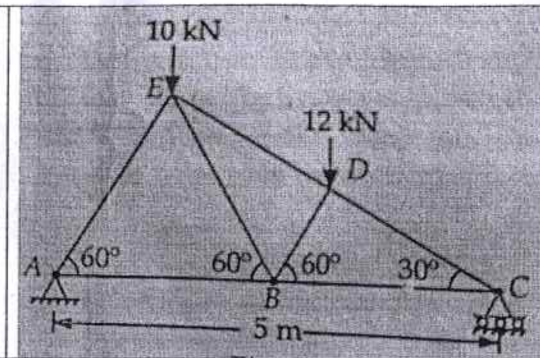


(c) Determine the horizontal and vertical centroidal moment of inertia of the sections shown in figure.



Q4 (20 marks) — CO4 CO5

(a) A frame is loaded and supported as shown in Figure. Determine the forces (magnitude and nature) in the all members.



(b) A body resting on a rough horizontal plane required a pull of 180 N inclined at 30^0 to the plane just to move it. It was found that a push of 220 N inclined at 30^0 to the plane just moved the body. Determine the weight of the body and the coefficient of friction.

(c) i. What is a frame. How it differes from a truss.
ii. State and explain the classification of a frame.
iii. State the assumptions for the static analysis of a frame.
(2+4+4)

Q5 (20 marks) — CO6

(a) i. Define the following terms
- Machine
- Mechanical Advantage
- Velocity ratio
- Reversible and irreversible machine
- Efficiency of a Machine

(5+5)

ii. Derive the expression for the condition of reversibility of a machine.

(b) i. What is law of a machine? Derive an equation for the same.
ii. In a lifting machine, an effort of 40 N raised a load of 1 kN. If efficiency of the machine is 0.5, what is its velocity ratio? If on this machine, an effort of 74 N raised a load of 2 kN, what is now the efficiency ? What will be the effort required to raise a load of 5 kN?

(c) Draw a neat sketch the following lifting machines and derive the expression of efficiency of the same.
i. Wheel and Differential Axle
ii. Screw Jack

(5+5)